

Part I

SUMMARY of Antiderivatives:

Summary

February 21, 2026

Summary

Suppose an object is traveling along a straight line, with displacement given by a function $s(t)$.

- The *average velocity*, v_{av} , of the object between time $t = a$ and time $t = b$ is given by

$$\begin{aligned}v_{av} &= \frac{\Delta s}{\Delta t} \\&= \frac{s(b) - s(a)}{b - a}\end{aligned}$$

- The *instantaneous velocity* of the object at time $t = a$ is given by

$$\begin{aligned}v(a) &= \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} \\&= \lim_{t \rightarrow a} \frac{s(t) - s(a)}{t - a} \\&= \lim_{t \rightarrow a} v_{av}(t)\end{aligned}$$

where $v_{av}(t)$ denotes the average velocity on the interval $[a, t]$ if $t > a$, and $[t, a]$ if $t < a$.

Part II

Recitation Questions

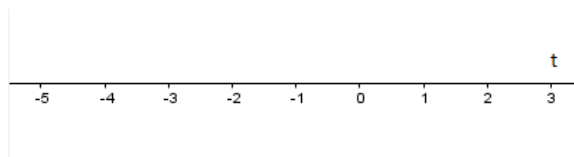
Problem 1

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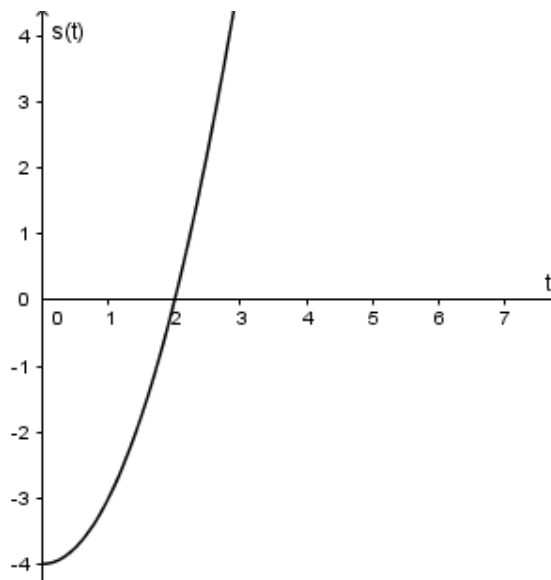
Problem 1

Problem 1 The position, $s(t)$, of an object moving along a horizontal line is given by $s(t) = t^2 - 4$, where s is in meters and t is in seconds, $0 \leq t < 5$.

- (a) Mark the position of the object on the line at time $t = 1$:



- (b) Find the average velocity, v_{AV} , of the object during the time interval $[1, 3]$.
- (c) Compute the average velocity, $v_{AV}(t)$, of the object during the time interval
- (i) $[1, t]$, for $1 < t < 5$;
 - (ii) $[t, 1]$, for $0 \leq t < 1$.
- (d) Find the instantaneous velocity, v_{inst} , of the object at $t = 1$. Justify your answer.
- (e) The position-time graph of the function s is given in the figure below.



- (i) Assume P is a point on the graph of s . Fill in the blank.

$$P = (1, \underline{\hspace{1cm}}).$$

Problem 1

- (ii) *Plot the point P and draw the tangent line at this point in the figure above.*
- (iii) *Find the slope, m_{tan} , of the tangent line in part (ii). Explain.*

Problem 2

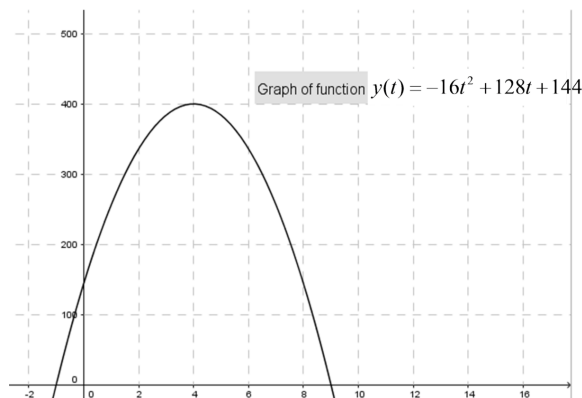
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Problem 2

Problem 2 Part of the given parabola can be used to model the “position-time” graph of a ball thrown straight up into the air. The graph gives the height of the ball in feet t seconds after being thrown into the air.

Let the function f be defined by $f(t) = -16t^2 + 128t + 144$.

Use this graph, and the given function, f , to answer the following questions.



- Mark the part of the parabola that can be used to model the position of the ball.
- What are the units on the t axis? What are the units on the y axis?
- If you were watching a movie of the ball being thrown, is the graph a picture of the path that the ball follows? Why or why not?
- Let $f(t)$ denote the height of the ball at time $t, t \geq 0$. What is the height of the ball at time $t = 0$?
- When will the ball hit the ground?
- What is the domain of the position function, f , of the ball?
- Use the table of values to find the average velocity of the ball between $t = 8.9$ and $t = 9$ seconds.

t	$\approx f(t)$
8.9	15.84
8.99	1.6
8.999	0.159984
8.9999	0.015998
9	0

Problem 2

- (h) Use the table of average velocities to approximate the instantaneous velocity of the ball when it hits the ground.

Time Interval	Average Velocity
$[8.9, 9]$	$\frac{f(9) - f(8.9)}{.1} = \frac{0 - 15.84}{.1} = -158.4$
$[8.99, 9]$	$\frac{f(9) - f(8.99)}{.01} = \frac{0 - 1.5984}{.01} = -159.84$
$[8.999, 9]$	$\frac{f(9) - f(8.999)}{.001} = \frac{0 - .159984}{.001} = -159.984$
$[8.9999, 9]$	$\frac{f(9) - f(8.9999)}{.0001} = \frac{0 - .0159998}{.0001} = -159.998$

- (i) Compute $v_{AV(t)}$ the average velocity of the ball on the time interval $[t, 9]$, where $t < 9$.
- (j) Compute $v(9)$, the **instantaneous** velocity of the ball at $t = 9$.